Received: 13 July 2023 | Revised: 11 August 2023 | Accepted: 24 September 2023

DOI: 10.1002/ima.22975

RESEARCH ARTICLE

WILEY

# Hybrid-NET: A fusion of DenseNet169 and advanced machine learning classifiers for enhanced brain tumor diagnosis

**Saif Ur Rehman Khan** | **Ming Zhao** | **Sohaib Asif** | **Xuehan Chen**

School of Computer Science and Engineering, Central South University, Changsha, China

**Correspondence**
Ming Zhao, School of Computer Science and Engineering, Central South University, Changsha, China.
Email: meanzhao@csu.edu.cn

**Funding information**
Natural Science Foundation of Hunan Province, China, Grant/Award Number: 2020JJ4757; New Generation Artificial Intelligence, Grant/Award Number: 2020AAA0109600; National Key R&D Program of China, Grant/Award Number: 2021ZD0140301; National Natural Science Foundation of China, Grant/Award Number: 61902433; High-Performance Computing Center of Central South University

**Abstract**

The computer-aided diagnostic (CAD) method to detect human brain tumors relies heavily on automated tumor characterization. Although CAD method has been extensively researched, significant obstacles still exist. Magnetic resonance imaging (MRI) classifiers brain tumors into glioma, meningioma, and pituitary tumors. However, accurately distinguishing between these tumor types remains a complex challenge in medical imaging. The recently developed deep learning and machine learning (ML) techniques have shown immense potential in image classification. However, the low numbers of medical image archives also pose a problem for medical image classification. As a result, fewer medical images are available to use in deep learning research and development. To address this issue, we use three highly effective ML classifiers with deep convolutional learned features for medical image classification. The MRI images of the three distinct types of brain tumors may be found in an open dataset on Figshare, which tests the automated approach's efficacy. Features are extracted from brain MRI scans using the DenseNet169 model. Extracted features are fed into a multiclass three ML classifiers (RF, SVM, XGBoost) for improved performance. Results from testing and evaluation of the entire framework are promising, especially compared to the field's state-of-the art technique. The suggested model outperformed the state-of-the-art technique, with an overall classification accuracy of 95.10%. To verify the enhanced performance of the proposed system, extensive tests are conducted on the brain MRI dataset available on Figshare. The optimal hyper-parameter fading classifier is seen to outperform the Softmax classifier for the features when there is limited training data.

**KEYWORDS**

brain tumor, DenseNet169, machine learning classifier, MRI, transfer learning

## 1 | INTRODUCTION

The computer-aided diagnostic (CAD) method has made significant progress in recent years due to advancements in imaging methods, image processing software, machine learning theories, and deep learning principles. In the field of medical image processing and analysis, research focuses on automatic disease identification, disease

 © 2023 Wiley Periodicals LLC. 

classification, and medical image retrieval for tracking disease progression. Various feature extraction and classification methods have been employed to address the challenge of disease detection and classification in medical images. However, determining the optimal characteristics and classifiers remains challenging, often relying on heuristics. Convolutional neural networks (CNNs), a type of deep learning architecture,[1] offer a promising solution by combining feature extraction and classification into a single computational module. CNNs have gained considerable attention and have demonstrated impressive results in CAD systems.

The brain, an intricate organ, regulates the nervous system and contains billions of cells. Tumors can infect it, causing structural changes and devastating neurological consequences. Brain tumors are categorized as meningioma, gliomas, and pituitary tumors based on cell type and growth location. Meningioma and gliomas are highly hazardous and can be fatal if not caught early. Gliomas are prevalent, accounting for 45% of cases,[2] while meningioma and pituitary tumors each make up 15%. Proper categorization is crucial for accurate diagnosis and treatment planning. The World Health Organization[3] classifies brain tumors into four stages, with stages 3 and 4 being the most severe. Accurate categorization is vital for effective management of brain tumors.

Medical image analysis is often the premier method for identifying and categorizing cancer. Magnetic resonance imaging (MRI),[2,4] X-rays,[5] and CT scans are only a few examples of medical image analysis techniques used to peer inside a human body. Among these imaging modalities, MRI is the gold standard for diagnosing different kinds of tumors.

MRI is a painless, non-invasive medical imaging technique that can generate high-quality images and provide invaluable 2D and 3D data on the kind and extent of a brain tumor. However, human evaluation of these images is difficult, error-prone, and time-consuming due to the high volume of individuals requiring screening. Furthermore, tumors can have a wide variety of shapes, which may make it difficult to recognize them. Automated and efficient CAD are needed to tackle this issue because they not only give precise evaluation and prompt diagnosis but also help keep costs down and act as a helpful tool for radiologists and doctors. There has been much focus on brain MRI investigations[2] recently. Efforts to establish a robust method for categorizing brain tumors have been made on multiple occasions.

Still, the varying shapes of MRI and contrast levels make this a complex problem. Recent developments in artificial intelligence have allowed the problem of detecting brain tumors to be solved by employing DL techniques. CNN is an effective method to automatically extract relevant characteristics from a given image without human involvement. In contrast to conventional machine learning techniques, CNN does not require any image pre-processing before classification.[4] These DL models are notoriously tedious and data-driven to train. However, a significant amount of data that is difficult to get is one of the critical challenges researchers encounter. Using pre-trained models on the ImageNet dataset has contributed to the rise in meta-learner popularity.

We present an automated MRI-based brain tumor classification method for glioma, meningioma, and pituitary tumors. Our motivation includes the need for improved precision in tumor classification, surpassing existing challenges, and overcoming limitations in medical image data. Our work contributes to the field by introducing an automated method that addresses these objectives.

- Our objective is to develop a deep hybrid voting convolutional network model called Hybrid-NET for accurate classification of brain tumors. The Hybrid-NET model combines multiple machine learning (ML) classifiers, including DenseNet169, support vector machine (SVM), XGBoost, and random forest (RF), to achieve high accuracy in diagnosing brain tumors.
- The Hybrid-NET model combines DenseNet169 with other classifiers like SVM, XGBoost, RF, and majority vote ensemble for tumor classification. It outperforms state-of-the-art algorithms in classification metrics, demonstrating superior performance.
- The Hybrid-NET model improves classification results through Grid Search for optimal hyperparameters and utilizing features from DenseNet169. Measures are taken to prevent overfitting by employing techniques that mitigate excessive memorization of the training data.

## 2 | RELATED WORK

DL-based strategies have been developed for brain tumor classification. Cheng created the Figshare dataset in 2015, consisting of brain MRI images, and updated it in 2017. They used Bag-of-Words (BoW),[2] GLCM, and an intensity histogram to identify three types of brain tumors, achieving 91.28% accuracy.[2] Another study[4] developed a deep CNN architecture with a small kernel, achieving 97.5% accuracy using texture and constitute features.[5] They employed a support vector machine classifier and tested their strategy on two datasets. In,[6] a hybrid approach combining PCA with normalized GISTs achieved a 94.23% success rate using a regularized extreme learning machine. Zhou et al.[7] used a

DenseNet-based ensemble and RNN method to classify brain tumors from MRI scans.

In this paper, we evaluate our proposed methodology using the Figshare dataset for brain tumor classification. Several studies have used this dataset to develop models with high accuracy. One study[8] achieved a 99.3% success rate using a CNN model based on the Transform of Discrete Wavelet. Another study[9] used a deep transfer learning model with GoogleNet architecture and SVM, achieving 98% accuracy. Kumar, Kakarla, Isunuri, and Singh[10] proposed a deep model based on pre-trained ResNet-50 with augmentation and transfer learning, achieving a 97.48% accuracy rate for classifying different types of brain tumors. Anaraki, Ayati, and Kazemi[11] presented a genetic algorithm-based seven-layer CNN model with 94.2% accuracy on the brain MRI dataset.

The detection of brain tumors using an ensemble model was proposed by Noreen, Palaniappan, Qayyum, Ahmad, and Alassafi.[12] They achieved a precision of 94.34%. Another study[13] achieved a performance of 97.39% using a four-layer CNN architecture for brain tumor classification. Çinar and Yildirim[14] achieved an accuracy of 97.2% in categorizing brain MR images using transfer learning (TL) and a modified version of ResNet-50. For brain state categorization with an MRI dataset,[15] suggested a ResNet34-based DL technique, utilizing TL and various augmentation strategies. The proposed model yielded excellent results.

Multiple studies have used pre-trained or meta-learner algorithms to classify brain tumors.[16] They achieved high accuracy, with one study reaching 97.37% accuracy by connecting two CNN models.[17] Automatic brain tumor segmentation using a CNN's multi-path architecture achieved 97.3% accuracy.[18] Fetal brain segmentation and prediction attained 89% accuracy with the Multi-Tasking Single Encoder model.[19] Brain MR image segmentation was proposed using a ResU-Net model, achieving 73.22% accuracy.[20] Experiments with five pre-trained models on two MRI datasets showed the Xception-based model performing at 99.67% accuracy.[21]

Most models struggle to achieve better classification recall and precision for brain tumors. Creating a stable model with higher accuracy is a challenging task. A more comprehensive version of the DCNN was proposed to address segmentation issues.[22] Multi-tumor occurrence is a widespread concern, leading to numerous studies.[23] Magnetic resonance and diffusion techniques are applied simultaneously, using a revolutionary multimodal super voxel-based segmentation technique called tensor imaging (DTI).[24] The results show quantitative improvements, with values at 86% and 0.84 for the clinical dataset, and 96% and 0.89 for the BRATS dataset, respectively.

The CAD framework[25] adopts a CNN-based methodology for tumor segmentation and classification. It comprises several stages, starting with image collection and resizing, followed by employing VGG-UNet for automated tumor segmentation and VGG16 for deep-feature extraction. The finest feature selection is accomplished through the application of the firefly algorithm, and binary classification is achieved via serial feature concatenation. Extensive experiments demonstrate the framework's efficacy, with SVM-Cubic achieving over 98% accuracy in a tenfold cross-validation for binary classification.

An automated system[26] for brain tumor classification utilizing deep learning in conjunction with an enhanced optimization algorithm. The system introduces a filtering fusion-based approach for tumor contrast enhancement and proposes a saliency map combined with active contour for tumor segmentation. It addresses challenges arising from the low contrast of original images and potential extraction of redundant features, which can result in incorrect tumor classification. Moreover, the authors offer a solution for multiclass classification, mitigating the issue of misclassification due to similar textural tumors.

This study[27] aims to develop a high-performance deep learning algorithm for accurate detection and classification of brain tumors in MR images. Leveraging the transfer learning model of AlexNet, to enhance the precision and dependability of radiology in brain tumor diagnosis. By automating the diagnostic process and optimizing radiologists' workflow, the algorithm seeks to detect and classify tumors at various stages and sizes, offering a potential solution to the escalating challenges in accurate brain tumor diagnosis.

The HHOCNN[28] method is proposed to enhance brain tumor classification. The method incorporates a candidate region process to accurately identify the tumor region while preserving essential edge details. Various features are extracted from the segmented region, which is then classified using CNN. The integration of the HHOCNN algorithm effectively minimizes the misclassification error rate and significantly improves overall tumor recognition accuracy. Notably, the HHOCNN achieves an impressive accuracy of 98% on the Kaggle dataset, showcasing its robust performance in brain tumor classification.

Maqsood, Damaševičius, and Maskeliūnas[29] propose a novel brain tumor detection and classification method utilizing a DNN and multiclass SVM. A customized 17-layered DNN architecture is developed for accurate brain tumor segmentation. Transfer learning is applied using a modified MobileNetV2 architecture for feature extraction and training. To ensure optimal feature

selection, an entropy-based controlled method in conjunction with multiclass SVM is employed. The proposed method achieves impressive accuracies of 97.47% for brain tumor detection and 98.92% for classification. The integration of Explainable AI (XAI) helps in interpreting the results.

We suggested a new deep hybrid voting convolutional network model named Hybrid-NET for the three-class classification problem, motivated by the achievements of the ensemble learning model[8,20,21,30] and the ML classifier. Brain tumors have been classified. This new Hybrid-NET model delivers remarkably higher accuracy than a single model for classifying different types of brain tumors.

## 3 | MATERIALS AND METHODS

The hybrid strategy in Figure 2 categorizes brain tumor diseases. The datasets used for training are described in Section 3.1. In Section 3.2, we preprocess MR images and use data augmentation to address limited datasets and improve classifier accuracy. Section 3.3 presents the Hybrid-NET model for feature extraction and categorization.

### 3.1 | Figshare dataset

On the Figshare dataset,[31–33] all previous efforts have been evaluated for their ability to distinguish between glioma, meningioma, and pituitary tumors. This is a publicly available dataset to investigate classification and retrieval issues in medical images. T1-weighted contrast-enhanced (CE) MRI slices from 233 patients comprising 3064 constitute the dataset. Many images of gliomas, meningiomas, and pituitary tumors are included in this unbalanced dataset, along with labels indicating which tumor type the 2D slices belong to. The dataset features multi-view slices in the axial, sagittal, and coronal planes. Downloadable in a.mat format, each "slice" is 512 pixels along the longest axis. Table 1 provides details on the dataset, and the following section displays some axial views of MRI scans included in the dataset.

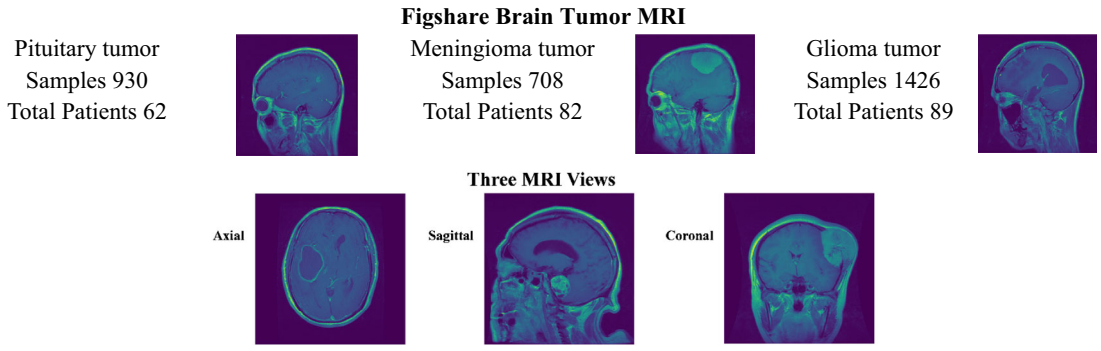
### 3.2 | Preprocessing and augmentation

The Figshare dataset underwent preprocessing to improve precision for generating the proposed architecture. This brain MRI dataset often contains blank areas and background noise. It is recommended to normalize the MR images to the same size. Different models have specific data requirements, such as DenseNet169 frameworks need images of size 224 by 224 pixels. All brain tumor images in this study were resized to 224 × 224 format. Chosen to resize the all images to 224 × 224 for DenseNet169 due to its compatibility with the architecture's input requirements, which often demand square-shaped images. Additionally, resizing images to the same size provides consistency, improves computational efficiency, aids generalization, and facilitates the use of pre-trained models, ultimately leading to better overall model performance.

Data splitting is crucial for image categorization. The image data was divided into training, testing, and validation sets using an 80/20 split ratio. This ratio is commonly used in medical image studies. Table 1 summarizes the distribution of images in the dataset, and Figure 1 shows samples of augmented MRI images.

In our study, we employed image augmentation techniques to address the limited number of images in the dataset and mitigate potential overfitting issues that could arise due to this scarcity. Instead of collecting new

**TABLE 1** Figshare brain tumor magnetic resonance imaging dataset summary.



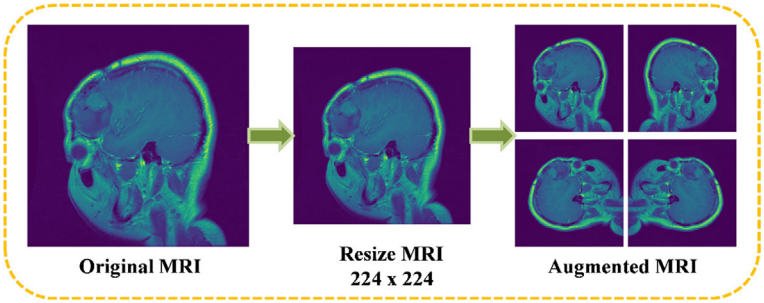



**FIGURE 1** Samples of augmented magnetic resonance images.

**TABLE 2** Dataset division detail.

| Figshare dataset | Dataset division | | |
|---|---|---|---|
| Multiclass (3) | Total MR images | Train set | Test set |
| Glioma (1426), Meningioma (708), and Pituitary tumor (930) | 3064 | 2451 | 613 |

data, we focused on enhancing the existing dataset to provide the DenseNet169 model with a more diverse and extensive training set. By doing so, we aimed to improve the classification performance of the deep learning model.

- 90° Rotation: Each image in the dataset was rotated by 90°, effectively creating additional variations of the original images. This augmentation helps the model become more robust to object orientations and improves its ability to recognize objects from different angles.
- MR Image Enlargement: We applied image enlargement to the MR images in the dataset, which increases the image's size while preserving its content. This technique introduces more variability in image scales and assists the model in handling objects of varying sizes.
- Flip: A horizontal flip was applied to the images, creating mirror-image versions of each original image. This augmentation allows the model to learn from horizontally flipped representations and encourages the recognition of objects from both left-to-right and right-to-left perspectives.

Table 2 provides detailed information about how the dataset was divided for training and testing purposes. By leveraging these augmentation strategies and optimizing the dataset division, we aimed to enhance the performance and generalization capabilities of the DenseNet169 model for our classification task.

## 3.3 | Methodology

The process of building a CNN from scratch is complex. To guarantee correct convergence of the model, this approach necessitates careful configuration of various hyperparameters (e.g., architecture depth), numerous passes, and an immense amount of knowledge. One alternative is using widely used pre-trained models (such as DenseNet169, etc.) to classify brain cancer. The current paucity of labeled data is a significant barrier to developing a robust and accurate detection method in medical imaging. Therefore, a TL technique or meta-learner is utilized to rapidly learn new tasks and address these challenges by leveraging previously trained models. Overfitting is mitigated, and the problem of insufficient training data is handled by employing deep transfer learning and data augmentation approaches in this research. A CNN trained for one task can be used for a different but related study in TL. The TL method proved faster and less complicated than a network trained from scratch. Pre-trained models such as EfficientNetB1, Xception, VGG16, ResNet50, and others are analyzed on the provided dataset of MR images. Compared to other pre-trained models, the DenseNet169 with ML classifier model fared the best.

To achieve this goal, we investigated various ML techniques using DenseNet169-derived features to assess the different performance evaluations of categorizing brain tumor lesions. We employed an ensemble method based on a majority vote to apply several machine learning techniques. We used the Grid Search technique to find the best hyperparameters for the Hybrid-NET architecture before turning to a majority vote. The following section provides a comprehensive breakdown of each stage.

### 3.3.1 | Pre-trained model

Transfer learning is a valuable technique for improving classification accuracy with smaller datasets. It leverages

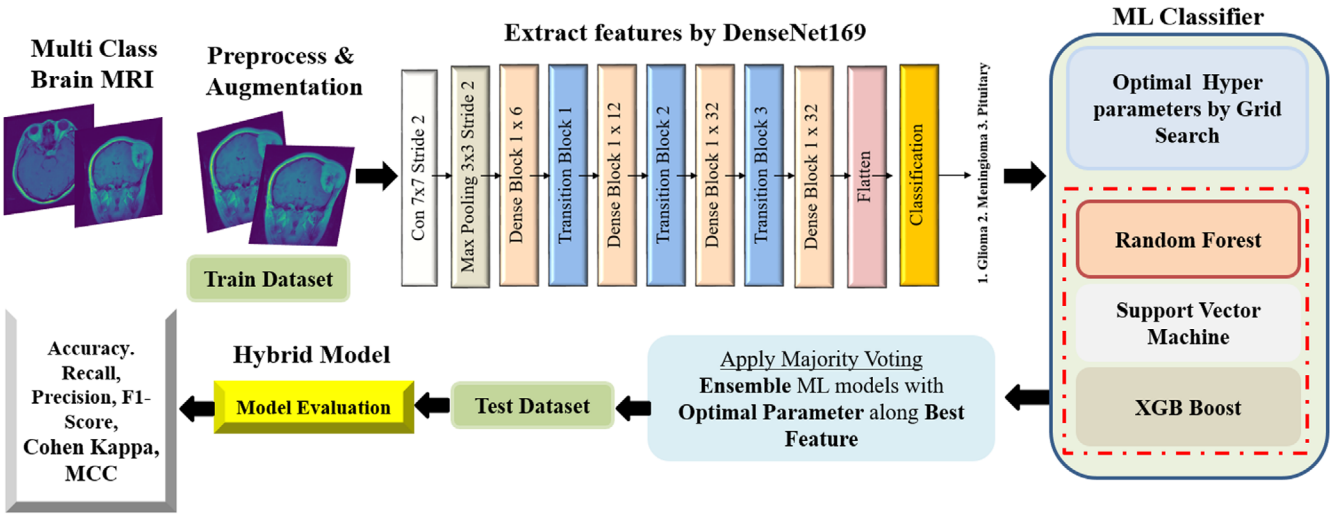



**FIGURE 2** Proposed Hybrid-NET architecture model for classification of brain lesion category.

models trained on larger benchmark datasets and has shown success in medical imaging applications.[27–29] Our research focuses on utilizing TL for brain image classification, benefiting from extensive training on diverse benchmark datasets to achieve accurate and reliable results with limited data.

We employ the DenseNet169 pre-trained CNN model for our research. DenseNet169 is a deep learning architecture pre-trained on a large-scale image classification task. By utilizing its extensive training, this model extracts intricate features and representations from images. With the DenseNet169 pre-trained model as our foundation, we can effectively classify brain images and identify brain-related patterns and abnormalities, benefiting from its captured general patterns and features.

*DenseNet169 model*

At the 2017 CVPR conference (Best Paper Award), the DenseNet was first proposed by Huang et al.[34] The study began with an attempt to build a deeper convolution network with shorter connections between its layers close to the input and those close to the output so that it would be easier and more effective to train. Unlike the ResNet, which relies on nonlinear transformation-skipping skip-connections, the DenseNet allows for direct connections between any two layers. Therefore, the feature maps of all previous layers ($x_0$ through $x_{i\text{-}1}$) are passed on to the $i$th layer as (1).[34]

$$x_i = H_i([x_0, x_1, x_2 \ldots\ldots x_{i-1}]) \quad (1)$$

where $x_0, x_1, x_2 \ldots\ldots x_{i-1}$ is the frequency distribution of the feature map generated in layers 0 through l-1. The robust block construction of five layers is depicted in Figure 2. Using the ImageNet pre-trained weights, the DenseNet169 model architecture[34] is illustrated in Figure 3.

### 3.3.2 | Machine learning classifiers

The data is cleaned and prepared before applying machine learning techniques for classification. Three classification algorithms—XGBoost, RF, and SVM—are evaluated. Strategies like Grid Search and feature selection techniques, including using DenseNet169 for feature extraction, are used to enhance the findings before applying the ML algorithms.

*Random forest*

Random forest has become increasingly popular in recent years for use in medical diagnostics.[35] The RF classifier constructs many trees of decisions rather than just one by randomly selecting seeds from the training set. It works for classification and regression but handles missing variables better for classification tasks. Each node in the forest examines the input and then votes on whether or not the category should be used to verify the items that came before it. At last, a random forest is used to choose the most popular option.

*Support vector machine*

By splitting the dataset in half, the hyperplane can be used using the supervised learning SVM. The second objective is to minimize faults while designing tasks in the first two categories (Classification and Regression). SVM's effectiveness persists even when dealing with larger sample sets. It is useful when the dimension is more significant than the sum of the samples. Additionally, SVM uses binary classification with two classes (healthy and heart illness symptoms).

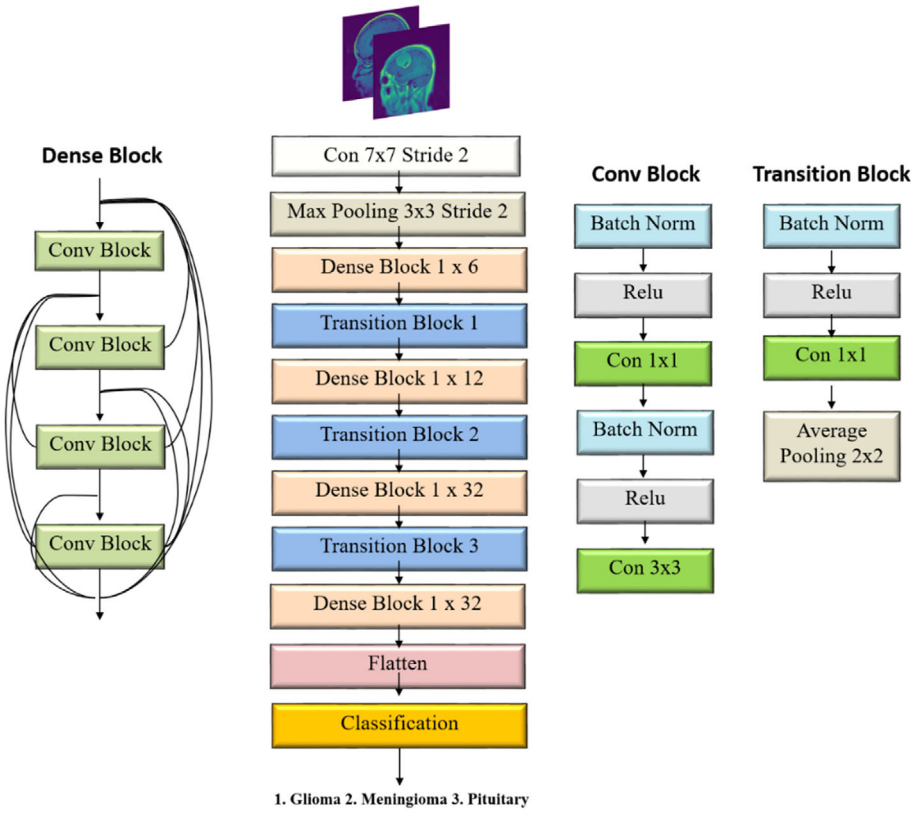



**FIGURE 3** Densenet169 basic architecture for tumor feature extraction.

*XGBoost*

The gradient boosting algorithm is implemented in XGBoost.[36] In various ways (memory use, processing in parallel, regularization, etc.), XGBoost outperforms GBM. It is a library explicitly made for gradient-enhanced tree models that are adaptable, fast, and powerful. Parameters and their values should be carefully considered when executing the XGBoost model. Parameter tuning is necessary for optimal performance when using the XGBoost model. It has advanced to being regarded as a "state-of-the-art" machine learning algorithm for handling structured data.

**TABLE 3** Optimal feature search by grid search.

| Methods | Optimal parameters |
|---|---|
| RF | Random stare: 30<br>Number of estimators: 500<br>Maximum depth: 8<br>Criterion: gini<br>Max features: auto |
| SVM | Gamma value: 0.0001<br>Kernel: Linear |
| XGBBoost | Number of estimators: 100<br>Learning rate: 0.02<br>Silent: true<br>Nthread:1<br>Objective: None |

### 3.3.3 | Grid search for optimal parameter

The traditional approach for hyperparameter optimization, known as a Grid Search, consists of a thorough investigation over a fixed region of the training algorithm's hyperparameter field. Since the parameter space for some parameters in the machine-learning approach may include areas with valid or finite values, we might need to set a boundary to employ a Grid Search. Although optimizing Grid Search in high-dimensional spaces can be challenging, the algorithm is usually easy to parallelize since its hyperparameter values are typically independent. Table 3 displays the optimum hyperparameter settings for the RF, SVM, and XGBoost methods.

### 3.3.4 | Hybrid architecture by integrating machine learning classifier with extracted features of DenseNet169 model

In this paper, we proposed a hybrid method to address the challenge of acquiring labeled data for training deep learning-based approaches. Our method uses a pre-

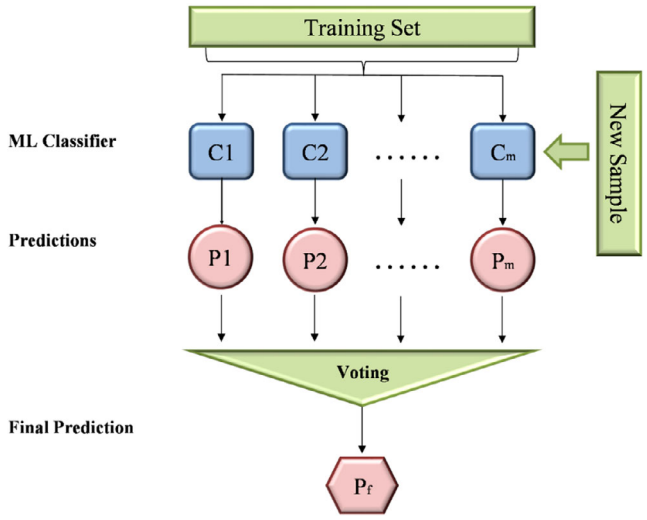



FIGURE 4 Voting basic architecture.

trained deep CNN called DenseNet169 to extract robust features from brain MR images. These features are then used with machine learning classifiers to determine if MR scans belong to the brain tumor group. We compared the features of eight models using seven machine learning classifiers and selected the best one using our feature evaluation and selection mechanism. Our method involves inputting the features into three different ML classifiers: random forest, support vector machine, and XGBoost.

In order to ensure the efficacy of our selected models. To achieve the highest performance possible, we conducted multiple evaluations with various model combinations. Through rigorous performance experimentation, we carefully selected and combined the models that we believe offer the most optimal performance compared to any other model selection. These chosen models have demonstrated superior capabilities and have been tailored to meet the specific requirements of our task.

### 3.3.5 | Ensemble hybrid architecture

Ensemble learning combines predictions from multiple algorithms to improve model performance (2). It is particularly useful in machine learning due to the diverse nature of methods. For our case, we use XGboost, SVM, and RF algorithms to accurately predict brain tumors. A hard vote strategy is employed for better precision, with each test case assigned a forecast and the outcome determined by majority vote. Figure 2 depicts the ensemble method, while Figure 4 shows the voting scheme architecture.

$$\hat{y} = mode\{C1(x), C2(x), ..., Cm(x)\} \tag{2}$$

## 4 | EXPERIMENTS AND RESULTS

Accuracy (3), precision (4), recall (5), and F1-score (6) were used to evaluate the models' ability to categorize brain tumors correctly. A confusion matrix is also introduced to visualize diagnostic examples of MR images generated by the proposed methodology. The following equations show how the suggested Hybrid-NET model was used to determine the overall performance of the trained model using test data.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{3}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5}$$

$$F1\,\text{Score} = 2 \times \frac{\text{Precion} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

True positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) are used in the equations mentioned above, respectively.

### 4.1 | Experimental setup

This section presents the methodology and findings of experiments conducted on Hybrid-NET architecture. It compares deep transfer learning with the ML classifier and the ensemble technique for the competitive model. The effectiveness of the suggested strategy is evaluated using the Figshare public brain tumor dataset, focusing on multiclass brain tumor classification accuracy. The proposed Hybrid-NET model is trained with Python and the Keras library, utilizing the GPU runtime type on Google Colaboratory Notebook. The model is trained using the NVIDIA GeForce MX350 with 12 GB of memory and 4 GB of RAM.

### 4.2 | Hyperparameters details and implementation

Details for the DenseNet169 (Base) model evaluation and their requirements for extracting multiclass features from brain tumors are listed in Table 4. This model uses an image size of 224 × 224 pixels for training and testing purposes. The number of epochs indicates how often the model iteratively processes the training dataset. All

TABLE 4 Hyperparameter detail (DenseNet169) model.

| Model | Image size | Epochs | Loss function | Optimizer | Activation | Learning rate | Batch size |
|---|---|---|---|---|---|---|---|
| DenseNet169 | 224 × 224 | 20 | Categorical | Adam | Softmax | 0.001 | 64 |

TABLE 5 Models for multiclass classification performance evaluation using a variety of metrics.

| Models | Sensitivity | Specificity | Precision | F1 Score | Accuracy |
|---|---|---|---|---|---|
| DensNet169 (Base) | 0. 88 | 0.85 | 0. 88 | 0. 88 | 88% |
| DenseNet169 + Random Forest | 0.91 | 0.90 | 0.91 | 0.90 | 91% |
| DenseNet169 + Support Vector Machine | 0.92 | 0.90 | 0.92 | 0.92 | 92% |
| DensNet169 + Extreme Gradient Boost | 0.94 | 0.93 | 0.94 | 0.94 | 94% |
| Hybrid-NET model | 0.95 | 0.94 | 0.95 | 0.94 | 95.10% |

models in this scenario are trained for 20 iterations. The optimization procedure is guided by the loss function, which evaluates the model's performance. Here, we focus on the loss function for classification tasks in which we have multiple classes to choose from, such as the glioma above, meningioma, and pituitary lesion. The optimizer changes the model's parameters during training to minimize the training loss. The chosen optimizer is Adam. Non-linearity is introduced into the model by fitting the activation function to the output of each neuron. In this situation, we employ the Softmax activation function. The optimizer makes small incremental changes to the model's parameters, and the learning rate controls this. In this case, the rate of learning is 0.001. The DenseNet169 model employs a batch size of 64. However, Table 3 provides specifics on the Grid Search optimizer's optimal hyperparameter search for the ML classifier.

## 4.3 | Performance analysis of proposed ensemble hybrid architecture

### 4.3.1 | Classification performance

In this research, we built five different models and compared their results using the criteria in Section 4. We provide a detailed analysis of the various architectural approaches. The primary goal of this research was to evaluate the effectiveness of the suggested Hybrid-NET model for the multiclass classification of brain tumors compared to the current state-of-the-art CNN model. The Figshare dataset was used in all experimental tests. The test results for all models on the test dataset are summarized in Table 5, including their accuracy, sensitivity, specificity, precision, and F1 score. The highlighted values in Table 5 indicate the optimal configuration for the specified performance indicators. As can be seen in Table 5, the test accuracy values across all models are relatively consistent. Hybrid-NET outperforms competing models across all categories, as demonstrated by the dataset. Table 5 shows that, generally, the Hybrid-NET model performs the best, with a 95.10% accuracy rate. Additionally, the model reaches the best possible values of sensitivity (0.95), specificity (0.95), accuracy (0.95), and F1 score (0.95). We believe in reducing the number of incorrect diagnoses of brain tumors, and the Hybrid-NET model's sensitivity was encouraging in this regard. The findings demonstrate that Hybrid-NET is superior in classifying different varieties of brain tumors. The deep convolutional model DensNet169 has outperformed other pre-trained models in our proposed Hybrid-NET. DenseNet[34] is a newly proposed CNN architecture that uses densely connected convolutional networks. Each layer in this network directly relates to all the layers below it. DenseNet is a condensed model that is easier to train and less likely to overfit.[37]

There are four substantial components. DenseNet's several benefits stem from the fact that all of its layers are directly exposed to the gradients of the loss functions, which helps fix the disappearing gradient problem, boosts the spread of parts, encourages reuse, and drastically lowers the number of parameters. The RF provides a unique combination of prediction accuracy and model interpretability among popular machine learning methods. RF's random sampling and ensemble strategies enable it to achieve accurate predictions and better generalizations. This generalization property comes from the bagging scheme, which improves the generalization by decreasing variance, while similar methods like boosting accomplish this by reducing bias.[38]

Regarding data modeling, SVMs are among the most effective methods. They use a method of controlling dimensionality known as generalization control. To facilitate comparisons between widely used model

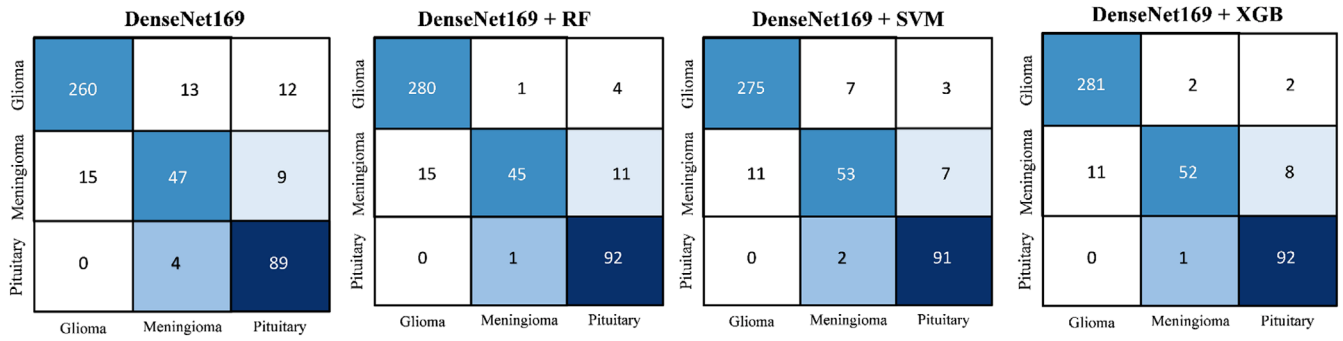



FIGURE 5 Confusion matric of DenseNet169, DenseNet169 + RF, DenseNet169 + SVM, and DenseNet169 + XGB.

architectures, kernel mapping is common ground. Maximizing the margin in classification issues is equivalent to minimizing the weight vector in a canonical framework, and both lead to greater control over the problem's generalization.[39] As a very effective scalable machine learning method for tree boosting, XGBoost has been used for various problems with excellent results. Significant systems and algorithmic optimizations, such as a novel tree learning algorithm, a theoretically justified weighted quantile sketch technique, and parallel and distributed computing,[36] contribute to XGBoost's scalability in all situations. A robust ensemble learning approach, tree boosting can combine multiple weak classifiers into a single robust one, improving classification accuracy. Ensemble learning combines predictions of various learning algorithms to boost a model's overall performance. Not all models can be used with data. When the multiple models are combined, the drawbacks of each are overcome. The accuracy of predictions has increased because of the majority vote. Each test case has its forecast that the majority vote decides.[30]

Classification results from the proposed Hybrid-NET ensemble model are shown in Figure 5 via confusion matrix analysis. The confusion matrix reveals that of the 449 images used in the test set, Hybrid-NET only made 22 incorrect classifications: four incorrect glioma classifications, 18 incorrect meningioma classifications, and zero incorrect pituitary classifications. Based on the findings of the evaluation metrics presented in Figure 6, it can be inferred that the Hybrid-NET model is superior to the Densenet169 + RF, DenseNet169 + SVM, DenseNet169 + XGB. Hybrid-NET has proven to be an excellent framework for the multiclass classification of MR images of brain tumors.

### 4.3.2 | Performance comparison of proposed ensemble hybrid architecture with existing methods

This section shows how our proposed Hybrid-NET ensemble model for identifying brain tumors compares

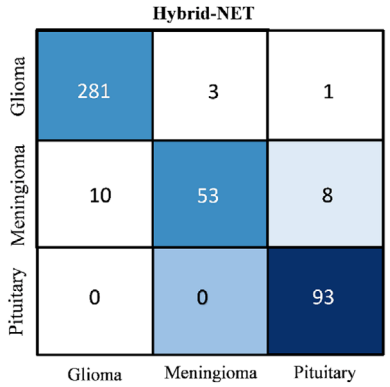



FIGURE 6 Proposed Hybrid-NET model confusion matrix.

TABLE 6 Performance comparison of proposed Hybrid-NET with advanced existing models.

| Reference | Method | Accuracy |
|---|---|---|
| Elhamzi et al.[40] | Hog with DSURF and ML-SVM | 90.02% |
| Landman et al.[41] | DNN with CNN | 91.43% |
| Palaniappan et al.[12] | Xception with KNN ensemble | 94.34% |
| Khan et al.[42] | Feature selection with DNN | 92.% |
| Pervaiz[43] | Modified AlexNet model | 91.16% |
| Hybrid-NET model | DensNet169 with majority vote ML | 95.10% |

significantly to existing DL approaches. Table 6 compares the suggested model's performance to that of previously reported models in brain tumor diagnostics. We considered the Figshare brain tumor dataset, which a significant proportion of previous studies used to evaluate model performance. Table 6 shows that our suggested ensemble model (majority voting) provides the highest accuracy compared to other current methods. Our ensemble Hybrid-NET model achieves extremely

outstanding outcomes because we systematically endeavor to construct a deep hybrid voting convolutional network model; this is the focus of this first-of-its-kind Hybrid-NET model (built on DenseNet169, support vector machine, extreme gradient boost, and random forest). A final result that is more accurate and generalizable than a single model is achieved by combining the predictions of the three ML classifiers using a majority vote technique. The final Flatten layer of the deep DenseNet169 model was mined for its deep feature sets. To carry out tumor classification, we used a set of classifiers that included the support vector machine, extreme gradient boost, random forest, and majority vote ensemble. The suggested Hybrid-Net model employs Grid Search with faded DenseNet169 features to locate the optimal hyperparameter, allowing for enhanced classification results. Overfitting can be avoided using several methods, including the Early Stopping Method.

#### 4.3.3 | Performance comparison of proposed ensemble hybrid architecture with deep learning models

Our approach was compared to nine pre-trained baseline models, Efficient Net (B1, B2), DenseNet121, InceptionV3, MobileNet, ResNet (50101), and VGG (16, 19), to highlight the benefits of the proposed model. For comparison, we test the base model using the same dataset. Classification accuracy, sensitivity, precision, and F1 score are reported in Table 7 between the pre-trained baseline model and the suggested model, with the best result highlighted in bold. Evidence that the DenseNet169 extractor features lead to the improved learning ability of the ML classifiers and that ensemble of these ML classifiers (majority voting) enhances the model's accuracy was higher than that of the nine pre-trained baseline models. Grid Search's suitable hyperparameter exhibits superior stability and a higher detection rate while classifying brain tumors. The method described in this paper is excellent for the baseline models that have already been trained.

#### 4.3.4 | Performance comparison of proposed ensemble hybrid architecture with machine learning classifier

Table 8 displays the final prediction accuracy of three classification algorithms trained with optimized hyperparameters, feature selection (DenseNet169) techniques, and an ensemble model (majority voting) strategy. Figure 5 shows the Hybrid-NET model's performance (Confusion Matrix) to that of the three classifiers. When compared to six different approaches to classification, the suggested model achieves a higher accuracy of 95.10% than any of them. Table 8 compares the methods' performance as measured by various standards, including their ability to classify data. The proposed model outperformed other models, with 95.10% accuracy, 95% recall, and 94% F1 score.

### 4.4 | Feature map analysis

Visual analysis of deep models is significant in real-world applications because it improves interpretability. Here, we look at how the suggested model uses the DenseNet169 model's visible and interpretable understanding to diagnose a specific region as a feature. To further comprehend how the DenseNet169 model functions, we display in Figure 7 a visualization of feature map analysis applied to MRI images of the brain. Visual inspection reveals that the model provides better interpretability and accurate detection of the diseased regions necessary for classification. In addition, Figure 7 depicts the DenseNet169 model's intermediate feature maps from various convolution blocks. We can observe how the model's input is broken down into various acquired filters using a feature map. Different features are extracted by the

**TABLE 7** Performance comparison of proposed Hybrid-NET with nine pre-trained baseline models.

| Models | Sensitivity | Specificity | Precision | F1 score | Accuracy |
|---|---|---|---|---|---|
| EfficientNetB3 | 0.90 | 0.88 | 0.89 | 0.89 | 90% |
| EfficientNetB5 | 0.83 | 0.81 | 0.88 | 0.85 | 83% |
| DenseNet201 | 0.84 | 0.83 | 0.85 | 0.81 | 84% |
| ResNet50 | 0.91 | 0.90 | 0.92 | 0.91 | 91% |
| RestNet101 | 0.92 | 0.90 | 0.91 | 0.91 | 92% |
| VGG16 | 0.89 | 0.86 | 0.88 | 0.88 | 89% |
| VGG19 | 0.89 | 0.88 | 0.89 | 0.89 | 89% |
| Hybrid-NET model | 0.95 | 0.94 | 0.95 | 0.94 | 95.10% |

**TABLE 8** Performance comparison of proposed Hybrid-NET with six machine learning classifiers.

| Models | Sensitivity | Specificity | Precision | F1 score | Accuracy |
|---|---|---|---|---|---|
| Decision tree | 0.85 | 0.84 | 0.86 | 0.85 | 85% |
| K nearest neighbor | 0.84 | 0.82 | 0.83 | 0.83 | 84% |
| Logistic regression | 0.85 | 0.84 | 0.87 | 0.86 | 85% |
| Naïve Bayes | 0.70 | 0.68 | 0.70 | 0.70 | 70% |
| Support vector machine | 0.88 | 0.86 | 0.89 | 0.88 | 88% |
| Random forest | 0.92 | 0.90 | 0.92 | 0.92 | 92% |
| Hybrid-NET model | 0.95 | 0.94 | 0.95 | 0.94 | 95.10% |

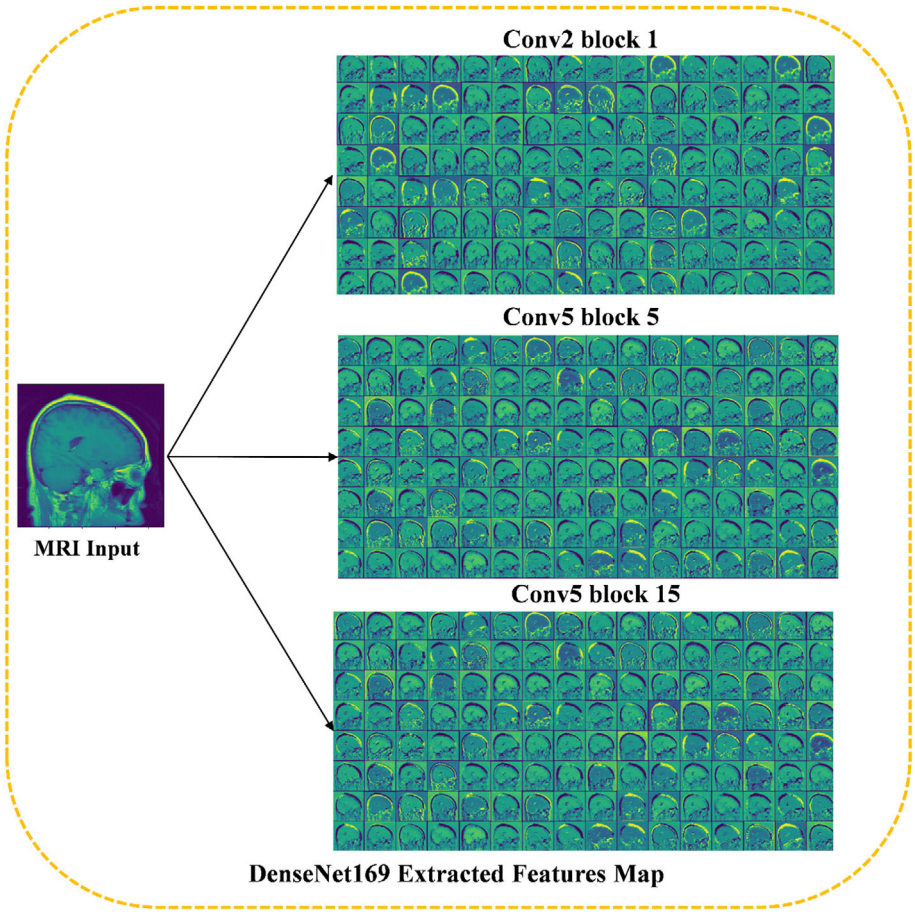



**FIGURE 7** Feature map extracted by DenseNet169.

Hybrid-NET model from the same input image, as is evident.

## 4.5 | Strength and future work

The results show that out of all the models tested, the DenseNet169 + XGBoost model performed the best in sensitivity, specificity, precision, F1 score, and accuracy. However, the Ensemble (majority vote) Architecture model outscored all other models and received the top scores in every category. These results indicate that the Hybrid-NET Architecture model, which integrates the predictions of different models, performs better than all of its components. The improved sensitivity, specificity, precision, F1 score, and accuracy demonstrated how it could classify brain tumors reliably. More study and validation are needed to verify the findings' generalizability to other datasets and clinical settings. The enhanced performance of the Ensemble Architecture model warrants further investigation, as does the possibility of further optimization for even better outcomes. Due to a lack of data about brain tumors, the scope of this investigation was restricted to a single institution. Better model

training necessitates higher-resolution MR images. Moreover, only T1W MRI data from a single protocol were employed in the analysis. Multiple MRI protocols can be used to make the system more reliable.

For future, we will plan to describe the sources of additional brain MRI data and the preprocessing steps required for data standardization and augmentation. Secondly, the experiment design will encompass data splitting, cross-validation protocols, and other details to ensure transparency and reproducibility. Additionally, we will acknowledge potential challenges with larger datasets, such as increased computational demands and data quality issues and outline strategies to address them effectively. Moreover, our aim is to improve the proposed model by exploring more sophisticated architectures, hyperparameter tuning, and considering state-of-the-art techniques in the field to enhance performance.

## 5 | CONCLUSION

This article concerns the multiclass classification of brain tumors and focuses on creating an automated Hybrid model for this purpose. Multimedia Tools and Applications multiclass categorization of brain tumors is complex because of the remarkable similarity between tumor kinds. The classification of brain tumors lacked the multiclass paradigm in earlier publications. Every image was resized to $224 \times 224$ in the preprocessing stage of our work utilizing image processing techniques, and we ran trials on the classification of three types of brain tumor patients. DenseNet169 is a deep transfer learning technology applied for feature extraction in brain tumor detection using magnetic resonance imaging after a performance study of deep learning architectures. The classification performance of the DenseNet169 model was enhanced by adding a flattened layer and a Softmax layer to the final layer. To detect brain tumor patients using MR images, we present a Hybrid-NET model based on the three ML classifiers (RF, SVM, XGBoost) architecture. Grid Search considerably enhances the classifier's performance with optimal hyperparameter search, and the suggested model's usage of the ensemble technique (majority vote) makes it very effective at learning several different high-level features. Using an ML classifier allows the proposed model to learn quickly, while data augmentation helps it avoid overfitting problems when extracting features.

Regarding overall classification accuracy, the suggested model reaches 95.10% on the Figshare dataset. The results show that the proposed technique outperforms the other models in this study on the target dataset. In terms of classification precision, our suggested model outperforms state-of-the-art alternatives. As a result, the proposed model can serve as a helpful tool for detecting and diagnosing various forms of brain tumors. We plan to enhance our study by experimenting with more significant amounts of brain MRI data in the future.

## ACKNOWLEDGMENTS

This work was supported by the Natural Science Foundation of Hunan Province, China (Grant No. 2020JJ4757). This work was supported by the Intelligent annotation and fine-grained recognition of large-scale multimodal medical behavior belong to 2030 Innovation Megaprojects (to be fully launched by 2020)-New Generation Artificial Intelligence (Project No. 2020AAA0109600). This work was funded by the National Key R&D Program of China under Grant No. 2021ZD0140301, the National Natural Science Foundation of China under Project No. 61902433, and the High-Performance Computing Center of Central South University.

## DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

## ORCID

*Saif Ur Rehman Khan* https://orcid.org/0000-0002-0768-5239

## REFERENCES

1. Asif S, Yi W, Ain QU, Hou J, Yi T, Si J. Improving effectiveness of different deep transfer learning-based models for detecting brain tumors from MR images. *IEEE Access*. 2022;10:34716-34730.
2. Kumar PS, Chatteijee S. Computer Aided Diagnostic for Cancer Detection Using MRI Images of Brain (Brain Tumor Detection and Classification System). *2016 IEEE Annual India Conference (INDICON)*. IEEE; 2016.
3. Rehman A, Naz S, Razzak MI, Akram F, Imran M. A deep learning-based framework for automatic brain tumors classification using transfer learning. *Circ Syst Signal Process*. 2020;39:757-775.
4. Seetha J, Raja SS. Brain tumor classification using convolutional neural networks. *Biomed Pharmacol J*. 2018;11(3):1457-1461.
5. Amin J, Sharif M, Yasmin M, Fernandes SL. A distinctive approach in brain tumor detection and classification using MRI. *Pattern Recogn Lett*. 2020;139:118-127.
6. Gumaei A, Hassan MM, Hassan MR, Alelaiwi A, Fortino G. A hybrid feature extraction method with regularized extreme learning machine for brain tumor classification. *IEEE Access*. 2019;7:36266-36273.
7. Zhou Y, Li Z, Zhu H, et al. Holistic brain tumor screening and classification based on densenet and recurrent neural network.

*Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 4th International Workshop, BrainLes 2018, Held in Conjunction with MICCAI 2018*, September 16, 2018, Revised Selected Papers, Part I 4. Springer; 2019.
8. Sarhan AM. Brain tumor classification in magnetic resonance images using deep learning and wavelet transform. *J Biomed Sci Eng*. 2020;13(6):102-112.
9. Deepak S, Ameer P. Brain tumor classification using deep CNN features via transfer learning. *Comput Biol Med*. 2019;111: 103345.
10. Kumar RL, Kakarla J, Isunuri BV, Singh M. Multi-class brain tumor classification using residual network and global average pooling. *Multimed Tools Appl*. 2021;80:13429-13438.
11. Anaraki AK, Ayati M, Kazemi F. Magnetic resonance imaging-based brain tumor grades classification and grading via convolutional neural networks and genetic algorithms. *Biocybern Biomed Eng*. 2019;39(1):63-74.
12. Noreen N, Palaniappan S, Qayyum A, Ahmad I, O. Alassafi M. Brain tumor classification based on fine-tuned models and the ensemble method. *Comput Mater Contin*. 2021;67(3):3967-3982.
13. Badža MM, Barjaktarović MČ. Classification of brain tumors from MRI images using a convolutional neural network. *Appl Sci*. 2020;10(6):1999.
14. Çinar A, Yildirim M. Detection of tumors on brain MRI images using the hybrid convolutional neural network architecture. *Med Hypotheses*. 2020;139:109684.
15. Talo M, Baloglu UB, Yıldırım Ö, Rajendra Acharya U. Application of deep transfer learning for automated brain abnormality classification using MR images. *Cogn Syst Res*. 2019;54:176-188.
16. Swati ZNK, Zhao Q, Kabir M, et al. Brain tumor classification for MR images using transfer learning and fine-tuning. *Comput Med Imaging Graph*. 2019;75:34-46.
17. Alshayeji M, al-Buloushi J, Ashkanani A, Abed S'. Enhanced brain tumor classification using an optimized multi-layered convolutional neural network architecture. *Multimed Tools Appl*. 2021;80(19):28897-28917.
18. Díaz-Pernas FJ, Martínez-Zarzuela M, Antón-Rodríguez M, González-Ortega D. A deep learning approach for brain tumor classification and segmentation using a multiscale convolutional neural network. *Healthcare*. Vol 9. MDPI; 2021.
19. Gangopadhyay T, Halder S, Dasgupta P, et al. MTSE U-net: an architecture for segmentation, and prediction of fetal brain and gestational age from MRI of brain. *Netw Model Anal Health Inform Bioinform*. 2022;11(1):50.
20. Anand I, Negi H, Kumar D, Mittal M, Kim TH, Roy S. Residual u-network for breast tumor segmentation from magnetic resonance images. *Comput Mater Contin*. 2021;67:3107-3127.
21. Onan A, Korukoğlu S, Bulut H. Ensemble of keyword extraction methods and classifiers in text classification. *Expert Syst Appl*. 2016;57:232-247.
22. Ahmed KB, Hall LO, Goldgof DB, Liu R, Gatenby RA. Fine-tuning convolutional deep features for MRI based brain tumor classification. *Medical Imaging 2017: Computer-Aided Diagnosis*. SPIE; 2017.
23. Mohsen H, el-Dahshan ESA, el-Horbaty ESM, Salem ABM. Classification using deep learning neural networks for brain tumors. *Future Comput Inf J*. 2018;3(1):68-71.
24. Soltaninejad M, Yang G, Lambrou T, et al. Supervised learning based multimodal MRI brain tumour segmentation using texture features from supervoxels. *Comput Methods Prog Biomed*. 2018;157:69-84.
25. Rajinikanth V, Kadry S, Nam Y. Convolutional-neural-network assisted segmentation and SVM classification of brain tumor in clinical MRI slices. *Inf Technol Control*. 2021;50(2):342-356.
26. Khan MA, Khan A, Alhaisoni M, et al. Multimodal brain tumor detection and classification using deep saliency map and improved dragonfly optimization algorithm. *Int J Imaging Syst Technol*. 2023;33(2):572-587.
27. Badjie B, Ülker ED. A deep transfer learning based architecture for brain tumor classification using MR images. *Inf Technol Control*. 2022;51(2):332-344.
28. Kurdi SZ, Ali MH, Jaber MM, Saba T, Rehman A, Damaševičius R. Brain tumor classification using meta-heuristic optimized convolutional neural networks. *J Pers Med*. 2023;13(2):181.
29. Maqsood S, Damaševičius R, Maskeliūnas R. Multi-modal brain tumor detection using deep neural network and multi-class SVM. *Medicina*. 2022;58(8):1090.
30. Asif S, Wenhui Y, Tao Y, et al. An ensemble machine learning method for the prediction of heart disease. *2021 4th International Conference on Artificial Intelligence and Big Data (ICAIBD)*. IEEE; 2021.
31. Cheng J, Huang W, Cao S, et al. Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLoS One*. 2015;10(10):e0140381.
32. Asif S, Zhao M, Chen X, Zhu Y. BMRI-NET: a deep stacked ensemble model for multi-class brain tumor classification from MRI images. *Interdiscip Sci*. 2023;15(3):499-514.
33. Asif S, Zhao M, Tang F, Zhu Y. An enhanced deep learning method for multi-class brain tumor classification using deep transfer learning. *Multimed Tools Appl*. 2023;82:1-28.
34. Huang G, Liu Z, van der Maaten L. Densely connected convolutional networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE; 2017.
35. Campanella G, Rajanna AR, Corsale L, Schüffler PJ, Yagi Y, Fuchs TJ. Towards machine learned quality control: a benchmark for sharpness quantification in digital pathology. *Comput Med Imaging Graph*. 2018;65:142-151.
36. Chen T, Guestrin C. Xgboost: A Scalable Tree Boosting System. in Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining. 2016.
37. Zhu Y, Newsam S. Densenet for dense flow. *2017 IEEE International Conference on Image Processing (ICIP)*. IEEE; 2017.
38. Yang P, Hwa Yang Y, B. Zhou B, Y. Zomaya A. A review of ensemble methods in bioinformatics. *Curr Bioinforma*. 2010; 5(4):296-308.
39. Cristianini N, Shawe-Taylor J. *An Introduction to Support Vector Machines and Other Kernel-Based Learning Methods*. Cambridge university press; 2000.
40. Ayadi W, Charfi I, Elhamzi W, Atri M. Brain tumor classification based on hybrid approach. *Vis Comput*. 2022;38(1): 107-117.
41. Paul JS, Plassard AJ, Landman BA, et al. Deep learning for brain tumor classification. *Medical Imaging 2017: Biomedical*

*Applications in Molecular, Structural, and Functional Imaging*. SPIE; 2017.
42. Sharif M, Tanvir U, Munir EU, Khan MA, Yasmin M. Brain tumor segmentation and classification by improved binomial thresholding and multi-features selection. *J Ambient Intell Humaniz Comput*. 2018;1-20.
43. Khawaldeh S, Pervaiz U, Rafiq A, Alkhawaldeh R. Noninvasive grading of glioma tumor using magnetic resonance imaging with convolutional neural networks. *Appl Sci*. 2017;8(1):27.

**How to cite this article:** Khan SUR, Zhao M, Asif S, Chen X. Hybrid-NET: A fusion of DenseNet169 and advanced machine learning classifiers for enhanced brain tumor diagnosis. *Int J Imaging Syst Technol*. 2024;34(1):e22975. doi:10.1002/ima.22975